

Create one VPC in N. Virginia region.

Your VPCs (1) [Info](#) Last updated less than a minute ago [Actions](#) [Create VPC](#)

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption contr...
☐	Devops-vpc	vpc-06680baa56c2f25e1	Available	-	-

Creatae one VPC

Create two subnets: one public subnet and one private subnet.

Subnets (1) [Info](#) Last updated 1 minute ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

Subnet ID : [subnet-0bcac1b4b444c2bec](#) [Clear filters](#)

☐	Name	Subnet ID	State	VPC
☐	Public-subnet	subnet-0bcac1b4b444c2bec	Available	vpc-06680baa56c2f25e1 Devo...

Public-subnet created with CIDR range 10.0.1.0/24

Subnets (1) [Info](#) Last updated less than a minute ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

Subnet ID : [subnet-0d86f6b29d1076e39](#) [Clear filters](#)

☐	Name	Subnet ID	State	VPC
☐	Private-subnet	subnet-0d86f6b29d1076e39	Available	vpc-06680baa56c2f25e1 Devo...

Private-subnet created with CIDR range 10.0.2.0/24

Attach an IGW to the VPC.

Internet gateway [igw-0faa26e69b97e50d4](#) successfully attached to [vpc-06680baa56c2f25e1](#)

igw-0faa26e69b97e50d4 / Devops-IGW [Actions](#)

Details [Info](#)

Internet gateway ID igw-0faa26e69b97e50d4	State Attached	VPC ID vpc-06680baa56c2f25e1 Devops-vpc	Owner 469132170253
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Attach the VPC(Devops-vpc) to the IGW

Create one public route table (RT) and one private route table.

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

Buttons: Add route, Remove, Cancel, Preview, Save changes

Add route by 0.0.0.0/0, choose internet gateway

Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources

Routes (2)

Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-Ofaa26e69b97e5...	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

So we attached IGW to our VPC, now we should manually give IGW to the public subnet

Available subnets (1/2)

Filter subnet associations

1

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Public-subnet	subnet-0bcac1b4b444c2bec	10.0.1.0/24	-	Main (rtb-0cb21451696b17968)
Private-subnet	subnet-0d86f6b29d1076e39	10.0.2.0/24	-	Main (rtb-0cb21451696b17968)

Choose the public subnet and save changes, so then the public subnet has IGW now

Route table rtb-091b376a4ac947bbe | Private-RT was created successfully.

rtb-091b376a4ac947bbe / Private-RT

Actions

Details Info

Route table ID

rtb-091b376a4ac947bbe

VPC

vpc-06680baa56c2f25e1 | Devops-vpc

Main

No

Owner ID

469132170253

Explicit subnet associations

-

Edge associations

-

Now create private route table associated with VPC

Available subnets (1/2)

Filter subnet associations

1

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
Public-subnet	subnet-0bcac1b4b444c2bec	10.0.1.0/24	-	rtb-0cb21451696b17968 / Public-RT
Private-subnet	subnet-0d86f6b29d1076e39	10.0.2.0/24	-	Main (rtb-0cb21451696b17968 / Pu

Selected subnets

subnet-0d86f6b29d1076e39 / Private-subnet

So for private-subnet is associated with private-RT

Deploy a NAT gateway in the public subnet and attach the NAT gateway to the private subnet.

✓ NAT gateway nat-127e961fc9bc29f23 | Devops-NAT was created successfully.

NAT gateways (1) Info

Find NAT gateways by attribute or tag

Name

NAT gateway ID

Connectivity...

State

State message

Availabi

Devops-NAT

[nat-127e961fc9bc29f23](#)

Public

Pending

-

Regiona

Created NAT-Gateway to the public subnet

Edit routes: rtb-091b376a4ac947bbe / Private-RT

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	✓ Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute

Add route

Cancel

Preview

Save changes

We need to connect NAT-Gateway to the private route table

Instances (1) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Filter Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	Public-EC2	i-05615be1e26379c51	✓ Running	t3.micro	⌚ Initializing	us-east-1a

Created a instance with public subnet

Instances (2) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Filter Instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	Public-EC2	i-05615be1e26379c51	✓ Running	t3.micro	✓ 3/3 checks passec	us-east-1a
☐	Private-EC2	i-02fc04a65afc1d5af	✓ Running	t3.micro	⌚ Initializing	us-east-1a

Created a EC2 using the private subnet with is connected to NAT Gateway

Deploy Apache server on both EC2 instances with a sample index.html file.

```

aws  [Alt+S]
#_
~\#### Amazon Linux 2023
~~\#####\
~~\####|
~~\##/ https://aws.amazon.com/linux/amazon-linux-2023
~~V~' '->
~~~
~~~.~.~
~/m/' '->
[ec2-user@ip-10-0-1-169 ~]$ sudo su -
[root@ip-10-0-1-169 ~]#

```

```
[root@ip-10-0-1-169 ~]# cat /var/www/html
cat: /var/www/html: Is a directory
[root@ip-10-0-1-169 ~]# cd /var/www/html
[root@ip-10-0-1-169 html]# ls
[root@ip-10-0-1-169 html]# touch index.html
[root@ip-10-0-1-169 html]# vi index.html
[root@ip-10-0-1-169 html]# curl localhost
<h1>Hello this is Jagan, from public EC2</h1>
```

Create one application load balancer and attach it to both EC2 instances.

☐	Name	Subnet ID	State	VPC
☐	public-subnet-2	subnet-01db2f9f60206329a	Available	ypc-06680baa56c2f25e1 Devo...

Created a Public-subnet 2

✓ You have successfully updated subnet associations for rtb-0cb21451696b17968 / Public-RT.

Available subnets (3/4)

☐	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	Public-subnet	subnet-0bcac1b4b444c2bec	10.0.1.0/24	–	rtb-0cb21451696b17968 / Public-
☐	Private-subnet	subnet-0d86f6b29d1076e39	10.0.2.0/24	–	rtb-091b376a4ac947bbe / Private-
☒	public-subnet-2	subnet-01db2f9f60206329a	10.0.3.0/24	–	rtb-0cb21451696b17968 / Public-
☒	public-subnet-3	subnet-0c5ba3fcfe377b	10.0.4.0/24	–	Main (rtb-0cb21451696b17968 / P

Now all the three subnets are associated with public subnet

Availability Zones and subnets | [info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-east-1a (use1-az1)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL Inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

[subnet-0bcac1b4b444c2bec \(Public-subnet\)](#)

10.0.1.0/24

Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (-)

☒ **us-east-1b (use1-az2)**

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL Inbound and outbound ALLOW rules, and internet gateway (IGW) routes configured to allow traffic.

[subnet-01db2f9f60206329a \(public-subnet-2\)](#)

10.0.3.0/24

Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (-)

Review targets

Targets (2) [Remove all pending](#)

☐ Show only pending < 1 > [Settings](#)

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4
i-02fc04a65afcd1d5af	Private-EC2	80	Running	launch-wizard-2	us-east-1a	10.0.2.138
i-05615be1e26379c51	Public-EC2	80	Running	launch-wizard-1	us-east-1a	10.0.1.169

Add both the instances to the target group

Hello this is Jagan, from public EC2

The ALB is currently serving the public-EC2 response.

Store application load balancer logs in S3.

Load balancers (1/1) [What's new?](#) [Actions](#) [Create load balancer](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

☒	Name	State	Type	Scheme	IP address type	VPC ID
☒	my-ALB	Active	application	Internet-facing	IPv4	vpc-0668

Load balancer: my-ALB

Logging - by Amazon CloudWatch [info](#)

Optimizes: Monitoring

Log delivery (0) [Edit](#) [Delete](#) [Add](#)

Load balancer: my-ALB

	Destination	Type	Log type	Delivery ID
☐	jagan-s3-...	S3	ALB_ACCE...	n3mnEM5ljSXbB9T3

Added s3 bucket to my-ALB

Store the VPC flow logs in a CloudWatch log group.

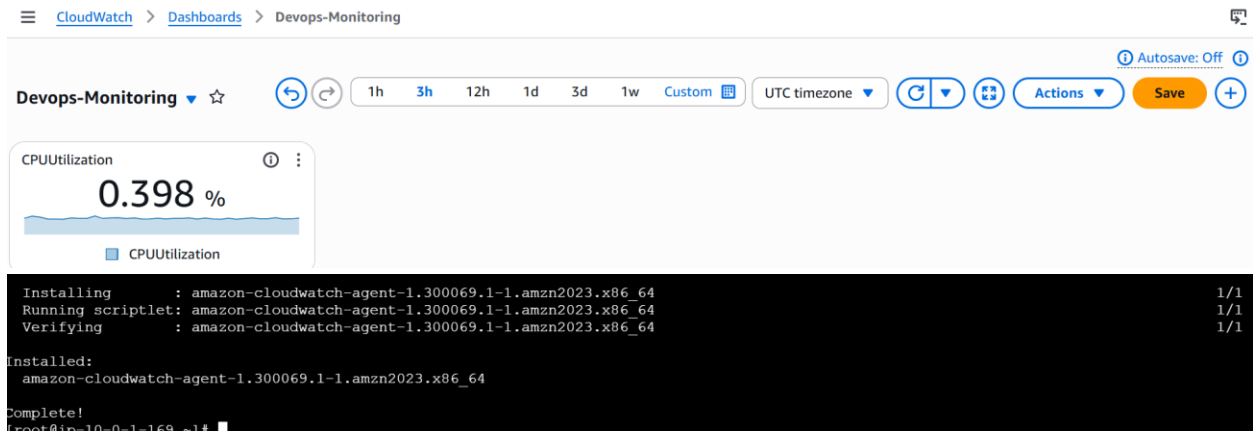
✔ Successfully created flow log for the following resource:
◦ vpc-06680baa56c2f25e1

fl-0f31b6115a8814bb1 / Devops-Flowlogs [Actions](#)

Details

Flow Log ID fl-0f31b6115a8814bb1	Destination Type cloud-watch-logs	Traffic Type All	File Format -
Name Devops-Flowlogs	Destination Name my-vpc-flow-logs	Max Aggregation Interval 10 minutes	Hive Compatible Partitions -
State Active	IAM Role arn:aws:iam::469132170253:role/service-role/VPCFlowLogs-Cloudwatch-1789385554832	Log Format Default	Partition Logs -

Create monitoring dashboards to monitor CPU utilization and to monitor the Apache service.



installed cloudwatch agent using (yum install amazon-clouswatch-agent -y)

```
{  
  
  "agent": {  
  
    "metrics_collection_interval": 60,  
  
    "run_as_user": "root"  
  
  },  
  
  "metrics": {  
  
    "namespace": "Custom/Apache",  
  
    "metrics_collected": {  
  
      "procstat": [  
  
        {  
  
          "pattern": "httpd",  
  
          "measurement": [  
  
            "pid_count"  
  
          ],  
  
          "metrics_collection_interval": 60  
  
        }  
  
      ]  
  
    }  
  
  }  
}
```

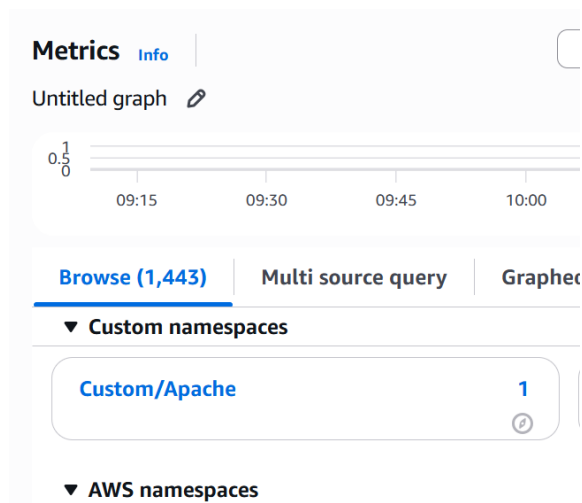
```
]
}
}
}
```

Enter the exact script

```
[root@ip-10-0-1-169 ~]# systemctl status amazon-cloudwatch-agent
● amazon-cloudwatch-agent.service - Amazon CloudWatch Agent
   Loaded: loaded (/etc/systemd/system/amazon-cloudwatch-agent.service; enabled; preset: disabled)
   Active: active (running) since Mon 2026-09-14 11:47:51 UTC; 47s ago
     Main PID: 79904 (amazon-cloudwat)
        Tasks: 7 (limit: 1059)
      Memory: 48.0M
         CPU: 407ms
    CGroup: /system.slice/amazon-cloudwatch-agent.service
            └─79904 /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent -config /opt/aws/amazon-cloudwatch-agent/etc/amazon-clou

Sep 14 11:47:51 ip-10-0-1-169.ec2.internal start-amazon-cloudwatch-agent[79909]: Executing /opt/aws/amazon-cloudwatch-agent/bin/amazon-clou
Sep 14 11:47:51 ip-10-0-1-169.ec2.internal start-amazon-cloudwatch-agent[79914]: D! [EC2] Found active network interface
Sep 14 11:47:51 ip-10-0-1-169.ec2.internal start-amazon-cloudwatch-agent[79914]: I! imds retry client will retry 1 timesI! Detected the in
Sep 14 11:47:51 ip-10-0-1-169.ec2.internal start-amazon-cloudwatch-agent[79914]: 2026/09/14 11:47:51 Reading json config file path: /opt/a
```

Cloudwatch agent is running



We can see our custom Apache

This means cloudwatch is successfully monitoring the Apache process

CloudWatch > Dashboards > Devops-Monitoring

You must choose Save Dashboard to save your changes to your custom dashboard. When you add a widget to a dashboard, it isn't saved automatically. [Save dashboard](#)

Autosave: Off

Devops-Monitoring ☆ 1h 3h 12h 1d 3d 1w Custom UTC timezone Actions Save

Apache Service Status

6

procstat_lookup_pid_count

If CPU utilization is more than 70%, then it should trigger auto scaling and launch new instance.

Devops-Launch-Template (lt-030fe61b3bbbd4a8a)

Actions

Delete template

Launch template details

Launch template ID

lt-030fe61b3bbbd4a8a

Launch template name

Devops-Launch-Template

Default version

1

Owner

arn:aws:iam::46913217025:root

Details

Versions

Template tags

Create a launch template first

Auto Scaling groups (1) Last updated less than a minute ago

Launch configurations Launch templates Actions Create Auto Scaling group

Search your Auto Scaling groups

☐	Name	Status - new	Launch template/configuration	Instances	Instance health - new	Desired
☐	Devops-ASG	At desired capacity	Devops-Launch-Template Version Default	1	1/1 Healthy	1

Created auto scaling group

EC2 > Auto Scaling groups > Devops-ASG > Dynamic scaling policy

Scaling policy name

Target Tracking Policy

Metric type

Average CPU utilization

Target value

70

Instance warmup

100 seconds

☐ Disable scale in to create only a scale-out policy

Cancel Create

Now the Autoscaling is done, when the load is heavy One EC2 is created and when the load is less EC2 gets deleted